

Assignment 1: Implement your own Linear Regression

Get the data [here](#)

Objective - Perform linear regression to make a model to find house prices.

1. Load and preprocess the dataset.
 2. Perform exploratory data analysis.
 3. Implement linear regression using the matrix-based method taught in class.
 4. Evaluate the performance of your regression model.
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Guidelines

- **Libraries you can use:** `numpy`, `pandas`, `matplotlib`, `seaborn` but **NOT** any ML or stat library
- Follow the matrix-based regression method taught in class.

Evaluate the Model

- Calculate performance metrics, including:
 - Mean Squared Error (MSE)
 - Residual sum of squares (RSS)
- Interpret your model's performance

Submission

- Submit a Jupyter Notebook with your code and explanations.
 - Include visualizations, outputs, and any observations.
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Bonus Task (Optional)

1. Experiment with standardising your independent variables. Standardization transforms input features with a mean of 0 and a standard deviation of 1.

For a feature x :

$$z = \frac{x - \mu}{\sigma}$$

Where:

- z : Standardized value.
- x : Original feature value.
- μ : Mean of the feature.
- σ : Standard deviation of the feature.

2. Extend your implementation to include polynomial regression by creating polynomial features.

A general polynomial regression equation:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \dots + \beta_n x^n + \epsilon$$

Submission Deadline

- Date: 31 May 2026 EOD
- Late submissions will incur a penalty.